SupportShift: A Theory-Linked Spatial Simulation Benchmark for Ignored Matérn Observation Support

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ABSTRACT

Spatial measurements often represent pixels, footprints, or local averages but are fitted as point observations. We introduce *SupportShift*, a parameterized synthetic benchmark that separates physical covariance parameters, population Kullback–Leibler (KL) targets, and finite-sample error under this support misspecification. Its analytic anchor is a stationary Matérn Gaussian field observed through a compact averaging kernel of bandwidth h. For fixed smoothness ν, ignoring support shifts the fitted inverse range at order $h^{2\nu}$ for $0 < \nu < 1$, $h^2 \log(1/h)$ at $\nu = 1$, and h^2 for $\nu > 1$, with explicit coefficients proving small-support range inflation. Because a variance-refitted two-point model is saturated, we extend the law to a weighted multi-lag composite and obtain both its information-weighted displacement and a strictly positive first nonzero residual-KL term. Under finite-design identifiability, a second result gives the Fisher-metric projection of the support perturbation onto the variance–decay tangent space and its irreducible full-likelihood KL component.

The released tracks test these claims by deterministic quadrature, exact finite-design targets, joint smoothness–range fitting, a partially corrected support model, matched-size boundary comparisons, and growing-p replicated fields. Joint fitting can reverse the fixed-smoothness decay direction, while increasing replication can concentrate a point-support estimator around a displaced target. A finite-library likelihood certificate permits arbitrary spatial dependence within each field. Code, synthetic generators, source tables, seeds, hashes, and pass/fail gates make every reported number auditable.

CCS CONCEPTS

• **Computing methodologies → Modeling and simulation**; **Model verification and validation**; • **Mathematics of computing** → *Probabilistic representations.*

KEYWORDS

Matérn covariance, change of support, Gaussian random field, high-dimensional probability, spatial simulation benchmark, misspecified likelihood

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1 INTRODUCTION

Values recorded on pixels, grid cells, or after kernel preprocessing are local aggregates of an underlying spatial field. Change-of-support models account for this by integrating the latent process over the measurement support [4, 10, 18, 21]. A common shortcut instead assigns each aggregate to a declared center and fits a point-support covariance. Relative to the latent parameters, the support is then misinterpreted. A single two-point covariance can still be matched exactly by different variance and decay parameters; a multi-lag likelihood is genuinely misspecified and selects a Kullback–Leibler (KL) projection [1].

This qualitative fact leaves quantitative and experimental questions. How quickly does the pseudo-parameter move as support shrinks? Does the rate depend on Matérn smoothness? Can directional support masquerade as directional range? Finally, does increasing dimension or replication remove the error, or merely make a wrong population target more precise? The first answer is not an ordinary second-order Taylor expansion: the Matérn covariance changes regularity at the origin as smoothness crosses one.

Block covariance and regularized variograms are classical [5, 10, 11]. Classical semivariogram regularization also shows that larger sampling support can increase an apparent or practical range. Clark's geometric rule concerns a finite range of influence, while Bellehumeur and Legendre use model-specific de-regularization approximations [2, 6]. A Matérn correlation has no finite range of influence; our target is instead the variance-refitted KL parameter at a fixed nonzero lag. Fuentes explicitly represents rectangular block averages in the spectral likelihood and invokes a small-pixel approximation [9]; the next-order point-fit parameter displacement is not part of that approximation. Modern aggregated-GP and INLA–SPDE methods likewise put the support operator inside the observation model [18, 21]. Convolution and covariance anisotropy can also be confounded [7, 16]. Generic misspecified-GP theory supplies the KL framework [1], while Gaussian quadratic-form concentration is standard [13, 15]. Table 1 separates these known ingredients from our claims. Pairwise and weighted composite likelihoods for Gaussian random fields are also established computational devices [3, 19]; our question is the local KL target created by an omitted observation operator, not composite-likelihood consistency or Godambe efficiency.

We contribute a theorem-linked benchmark rather than a new generic change-of-support method:

- We derive an all-smoothness, d-dimensional fixed-lag phase law with explicit coefficients, a universal small-support range direction, and a two-term approximation continuous through the smoothness-one threshold. We extend it to a genuinely misspecified multi-lag composite, including the first nonzero minimum-KL term, and to a finite-design full-Gaussian tangent projection.

- We derive a directional h^2 apparent-range contrast for elongated support. The qualitative link between convolution and anisotropy is known.
- We certify finite-library likelihood concentration and ERM excess risk for N independent p-dimensional Gaussian fields without assuming independent spatial coordinates.
- SupportShift turns the theory into reusable synthetic tracks spanning continuous and finite-design likelihoods, directional and boundary support, joint smoothness–range selection, and growing-p replicated fields. Each exposes declared scale controls, deterministic seeds, source tables, numerical gates, and a machine-checkable claim audit.

The fixed-smoothness restriction is substantive because information about Matérn smoothness can be both appreciable and design-dependent [8]. The general full-grid projection has a design-specific coefficient and no universal sign; joint smoothness can also reverse the fixed-smoothness decay direction. The high-dimensional experiment grows domain at fixed lattice spacing and uses independent replicated fields. We do not claim separate fixed-domain consistency of Matérn variance and range, which would conflict with microergodicity results [14, 20].

2 MODEL AND EXACT PAIR TARGET

Let $Y = \{Y(t) : t \in \mathbb{R}^d\}$ be a mean-zero stationary Gaussian field with

$$C(r) = \sigma^2 \mathcal{M}_\nu(\alpha\|r\|), \qquad \mathcal{M}_\nu(x) = \frac{2^{1-\nu}}{\Gamma(\nu)} x^\nu K_\nu(x), \tag{1}$$

where $\sigma^2, \alpha, \nu > 0$ and $\mathcal{M}_\nu(0) = 1$. Thus σ^2 is the marginal variance and α is inverse range. Let k be a symmetric probability density supported on $B(0, L)$, set $k_h(u) = h^{-d}k(u/h)$, and observe

$$Z_h(t) = \int k_h(u) Y(t-u)\, du. \tag{2}$$

For independent $U, V \sim k$, write $D = U - V$, $\Sigma_k = \mathbb{E}(UU^\top)$, $T_k = \operatorname{tr}(\Sigma_k)$, and $m_q = \mathbb{E}\|D\|^q$. Then $m_2 = 2T_k$, and stationarity gives

$$C_h(r) = \sigma^2 \mathbb{E}[\mathcal{M}_\nu\{\alpha\|r + hD\|\}]\,. \tag{3}$$

Assume $T_k > 0$.

We reserve η for a generic inverse-range candidate. A dagger denotes an exact population KL target under the point-support family, a tilde denotes an analytic approximation, and the superscript C denotes the weighted pair composite.

Fix $r = Re$, $R > 0$, and $\|e\| = 1$. A naive point-support pair model with known ν and parameters $\varsigma^2, \eta > 0$ uses

$$\varsigma^2 \begin{pmatrix} 1 & \mathcal{M}_\nu(\eta R) \\ \mathcal{M}_\nu(\eta R) & 1 \end{pmatrix}. \tag{4}$$

The smoothed two-by-two covariance lies in this family. Moreover, $0 < \rho_h(r) < 1$: positivity follows from $k \geq 0$ and $\mathcal{M}_\nu > 0$, while strict inequality follows from the positive smoothed spectral density near the origin. Hence the exact Gaussian KL target is

$$(\sigma_h^2)^\dagger = C_h(0), \qquad \mathcal{M}_\nu(\alpha_h^\dagger R) = \rho_h(r) := \frac{C_h(r)}{C_h(0)}. \tag{5}$$

It is unique because

$$\mathcal{M}_\nu'(x) = -\frac{2^{1-\nu}}{\Gamma(\nu)} x^\nu K_{\nu-1}(x) < 0. \tag{6}$$

A standard route to pair-composite consistency would use a fixed lattice, increasing rectangular Følner domains, a compact parameter space, target uniqueness and interiority, correlations bounded away from one, and a uniform ergodic law with an integrable envelope. We do not establish that uniform law. Dependence among overlapping pairs would require Godambe uncertainty, not an independence Hessian.

3 SMOOTHNESS-DEPENDENT PHASE LAW

Put $z = \alpha R$, $a_k(e) = e^\top \Sigma_k e$, and

$$B_{\nu,k}(r) = \alpha^2 \left[a_k(e)\mathcal{M}_\nu''(z) + \{T_k - a_k(e)\} \frac{\mathcal{M}_\nu'(z)}{z} \right]. \tag{7}$$

Proposition 1 (Fixed nonzero lag). *Uniformly for r in a compact annulus bounded away from zero, for $h \leq \|r\|/(4L)$, and for α in a compact subset of $(0, \infty)$,*

$$\frac{C_h(r)}{\sigma^2} = \mathcal{M}_\nu(\alpha\|r\|) + h^2 B_{\nu,k}(r) + O(h^4). \tag{8}$$

At zero lag, the Matérn origin expansion produces a phase transition.

Theorem 1 (Matérn support phase law). *Fix $\nu > 0$. Let*

$$c_\nu = \frac{\Gamma(1-\nu)}{\nu 2^{2\nu}\Gamma(\nu)} > 0, \qquad 0 < \nu < 1.$$

Under the preceding assumptions, as $h \downarrow 0$,

$$\frac{C_h(0)}{\sigma^2} = 1 - c_\nu m_{2\nu}(\alpha h)^{2\nu} + O(h^2), \qquad 0 < \nu < 1, \tag{9}$$

$$= 1 - \frac{m_2}{2}(\alpha h)^2 \log\{1/(\alpha h)\} + O(h^2), \qquad \nu = 1, \tag{10}$$

$$= 1 - \frac{m_2}{4(\nu-1)}(\alpha h)^2 + o(h^2), \qquad \nu > 1. \tag{11}$$

The pseudo-decay in (5) *satisfies*

$$\alpha_h^\dagger - \alpha = \frac{\mathcal{M}_\nu(z) c_\nu m_{2\nu} \alpha^{2\nu}}{R\mathcal{M}_\nu'(z)} h^{2\nu} + O\Big(h^{\min(4\nu,2)}\Big), \quad 0 < \nu < 1, \tag{12}$$

$$\alpha_h^\dagger - \alpha \sim \frac{m_2 \mathcal{M}_1(z)\alpha^2}{2R\mathcal{M}_1'(z)} h^2 \log(1/h), \qquad \nu = 1, \tag{13}$$

$$\alpha_h^\dagger - \alpha = \frac{A_{\nu,k}(r)}{R\mathcal{M}_\nu'(z)} h^2 + \mathcal{R}_\nu(h), \qquad \nu > 1, \tag{14}$$

where

$$A_{\nu,k}(r) = \alpha^2\{(2\nu-2)a_k(e) + T_k\} G_\nu(z) > 0, \tag{15}$$

$$G_\nu(z) = \frac{\mathcal{M}_\nu(z)}{2\nu-2} + \frac{\mathcal{M}_\nu'(z)}{z} \tag{16}$$

$$= \frac{2^{1-\nu}}{\Gamma(\nu)} \frac{z^\nu K_{\nu-2}(z)}{2\nu-2} > 0, \tag{17}$$

and $\mathcal{R}_\nu(h) = O(h^{2\nu})$ for $1 < \nu < 2$, $O\{h^4 \log(1/h)\}$ for $\nu = 2$, and $O(h^4)$ for $\nu > 2$. For fixed ν, the remainders are uniform over compact positive decay sets and lag annuli bounded away from zero, but not as ν approaches one or two. Consequently, for every $\nu > 0$ and all sufficiently small $h > 0$,

$$(\sigma_h^2)^\dagger < \sigma^2, \qquad \rho_h(r) > \mathcal{M}_\nu(\alpha R), \qquad \alpha_h^\dagger < \alpha.$$

Table 1: Nearest known ingredients and the narrower SupportShift addition.

Literature	Established result	Addition studied here
Change of support and block covariance [5, 9, 10, 18]	Integrate the latent covariance over known observation supports	Explicit small-support Matérn *point-fit pseudo-range* phase law
Regularized-semivariogram range rules [2, 6]	Support can enlarge a finite or practical variogram range	Infinite-range Matérn KL target, smoothness phase, and coefficient
Misspecified GP likelihood [1]	Population KL pseudo-target under covariance misspecification	Local tangent projection and irreducible KL for ignored support
Spatial composite likelihood [3, 19]	Pairwise objectives, weighting, consistency, and efficiency	Support-induced multi-lag target coefficient and residual KL
Convolution and anisotropy [7, 16]	Support/convolution can create anisotropic covariance	Leading directional apparent-range contrast
Gaussian quadratic forms [13, 15]	Nonasymptotic concentration of Gaussian quadratic forms	Direct certificate for the benchmark's structured covariance library
Spatial benchmarks [12]	Shared synthetic data and auditable method comparisons	Support-specific generator with analytic targets and pass/fail gates

The two-point family in (5) is saturated: after refitting variance, one correlation determines one decay exactly. To obtain a genuine misspecification result, let $0 < R_1 < \cdots < R_J$, $w_j > 0$, and $\mathrm{R}(\rho) = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$. Write $\rho_{h,j} = C_h(R_j e)/C_h(0)$ and define the Gaussian pair composite

$$Q_h^{\mathrm{C}}(\eta) = \sum_{j=1}^{J} w_j D_{\mathrm{KL}}\{\mathcal{N}(0, \mathrm{R}(\rho_{h,j})) \,\|\, \mathcal{N}(0, \mathrm{R}(\mathcal{M}_\nu(\eta R_j)))\}. \tag{18}$$

Define the regime-dependent scale

$$s_\nu(h) = \begin{cases} h^{2\nu}, & 0 < \nu < 1, \\ h^2 \log(1/h), & \nu = 1, \\ h^2, & \nu > 1. \end{cases} \tag{19}$$

Theorem 1 defines positive lag coefficients κ_j by $\alpha - \alpha_{h,j}^\dagger = \kappa_j s_\nu(h) + o\{s_\nu(h)\}$.

Theorem 2 (Genuinely misspecified multi-lag target). *Let α be interior to a compact positive parameter interval and let α_h^{C} minimize* (18) *on that interval. Put*

$$\rho_j = \mathcal{M}_\nu(\alpha R_j), \qquad \mathcal{I}_j = \frac{1 + \rho_j^2}{(1 - \rho_j^2)^2}, \tag{20}$$

$$\lambda_j = w_j \mathcal{I}_j \{R_j \mathcal{M}_\nu'(\alpha R_j)\}^2, \qquad \bar{\kappa} = \frac{\sum_j \lambda_j \kappa_j}{\sum_j \lambda_j}. \tag{21}$$

Then

$$\alpha - \alpha_h^{\mathrm{C}} = \bar{\kappa} s_\nu(h) + o\{s_\nu(h)\}, \tag{22}$$

$$\min_\eta Q_h^{\mathrm{C}}(\eta) = \frac{s_\nu(h)^2}{2} \sum_{j=1}^{J} \lambda_j (\kappa_j - \bar{\kappa})^2 + o\{s_\nu(h)^2\}. \tag{23}$$

Thus the composite still has apparent range inflation, and its minimum KL is strictly positive to leading order whenever the lag-specific coefficients are not all equal.

The coefficient is explicitly lag-sensitive through both κ_j and the Gaussian correlation information $\mathcal{I}_j$. It is neither the coefficient at a representative lag nor an unweighted average. This answers the pairwise versus multi-location objection without treating overlapping pairs as independent for uncertainty quantification.

The leading law is pointwise in fixed ν. To resolve its slow crossover, let $b_\nu = \Gamma(-\nu)/\{2^{2\nu}\Gamma(\nu)\}$ and define, for $0 < \nu < 2$,

$$W_{\nu,k}(h) = \begin{cases} \dfrac{m_2(\alpha h)^2}{4(1-\nu)} + b_\nu m_{2\nu} (\alpha h)^{2\nu}, & \nu \neq 1, \\ \dfrac{(\alpha h)^2}{2} \left[m_2 \{\log(\alpha h/2) + \gamma_{\mathrm{E}} - 1/2\} + \ell_{2,k} \right], & \nu = 1, \end{cases} \tag{24}$$

where $\ell_{2,k} = \mathbb{E}\{\|D\|^2 \log\|D\|\}$. Set

$$\widetilde{\rho}_h = \frac{\mathcal{M}_\nu(z) + h^2 B_{\nu,k}(r)}{1 + W_{\nu,k}(h)}, \qquad \mathcal{M}_\nu(\widetilde{\alpha}_h R) = \widetilde{\rho}_h. \tag{25}$$

Proposition 2 (Transition-aware approximation). *For fixed $0 < \nu < 2$,*

$$\alpha_h^\dagger - \widetilde{\alpha}_h = \begin{cases} O(h^{2\nu+2}), & 0 < \nu < 1, \\ O\{h^4 \log(1/h)\}, & \nu = 1, \\ O(h^4), & 1 < \nu < 2. \end{cases} \tag{26}$$

Moreover $W_{\nu,k}(h) \to W_{1,k}(h)$ as $\nu \to 1$ for fixed $h > 0$. The error orders are not claimed jointly uniform in (ν, h).

Proposition 3 (Directional support contrast). *Let $\Delta_e(h) = \alpha - \alpha_h^\dagger(e)$. For two unit lag directions e_1, e_2, under the theorem assumptions,*

$$\Delta_{e_1}(h) - \Delta_{e_2}(h) = \frac{\alpha^2 \{a_k(e_1) - a_k(e_2)\}}{R} \frac{K_{\nu-2}(z)}{K_{\nu-1}(z)} h^2 + o(h^2). \tag{27}$$

The formula holds for every $\nu > 0$. When $\nu \leq 1$, it is lower order than the direction-independent leading shift in the phase theorem.

For the benchmark, let

$$Z_{h,A}(t) = \int k(u) Y(t - hAu)\, du.$$

Then $\Sigma_k = AA^\top/5$. We vary the principal singular-value aspect ratio $\varrho \geq 1$ while holding $\mathrm{tr}(\Sigma_k)$ fixed using

$$A_\varrho = \sqrt{\frac{2}{\varrho + \varrho^{-1}}}\,\mathrm{diag}(\sqrt{\varrho}, \varrho^{-1/2}). \tag{28}$$

For $\nu > 1$, the ratio of the major- to minor-axis leading shift coefficients is

$$\frac{(2\nu - 1)\varrho^2 + 1}{\varrho^2 + 2\nu - 1}. \tag{29}$$

Thus elongated support can induce direction-dependent inferred range even for an isotropic latent covariance.

Remark 1. The fitted lag R is fixed and bounded away from zero. If $R = ch$, the nonzero-lag expansion is no longer uniform and the relative displacement need not vanish.

Thus ignored support inflates inferred range. The sign is not an assumption: for $\nu > 1$, it follows from positivity of $K_{\nu-2}$ in the displayed identity. For an isotropic kernel, $A_{\nu,k}(r) = \sigma_k^2\alpha^2(2\nu + d - 2)G_\nu(z)$.

3.1 Proof sketch

Let $f_\alpha(x) = \mathcal{M}_\nu(\alpha\|x\|)$. At a nonzero lag, fourth-order Taylor expansion is uniform. Symmetry and $\mathbb{E}(DD^\top) = 2\Sigma_k$ cancel odd terms, including the cubic term, and yield (8). At zero, the noninteger Bessel expansion is

$$\mathcal{M}_\nu(x) = \sum_{j\geq 0}\frac{(x^2/4)^j}{j!(1-\nu)_j} + \frac{\Gamma(-\nu)x^{2\nu}}{2^{2\nu}\Gamma(\nu)}\sum_{j\geq 0}\frac{(x^2/4)^j}{j!(1+\nu)_j}. \tag{30}$$

Taking expectations at $x = \alpha h\|D\|$ proves (9) and (11). At $\nu = 1$,

$$\mathcal{M}_1(x) = 1 + \frac{x^2}{2}\{\log(x/2) + \gamma_{\mathrm{E}} - 1/2\} + O\{x^4(1 + |\log x|)\},$$

which proves (10). Divide (8) by the relevant variance expansion and expand the inverse of $\alpha \mapsto \mathcal{M}_\nu(\alpha R)$. This gives (12)–(14). For $\nu > 1$, the Matérn differential equation and the recurrence

$$K_\nu(z) = K_{\nu-2}(z) + 2(\nu - 1)K_{\nu-1}(z)/z$$

reduce the normalized-correlation coefficient to $A_{\nu,k}(r)$, proving its sign. Appendix A records the proof details used by these claims; the reproducible technical manuscript also gives an exact exponential benchmark.

For (27), subtract the two directional fixed-lag expansions; the common zero-lag normalization and all direction-free terms cancel. Using

$$\mathcal{M}_\nu''(z) - \mathcal{M}_\nu'(z)/z = \frac{2^{1-\nu}}{\Gamma(\nu)}z^\nu K_{\nu-2}(z)$$

and dividing by $-R\mathcal{M}_\nu'(z)$ gives the displayed coefficient.

For Theorem 2, the KL divergence between unit-variance Gaussian pairs with correlations ρ and q has second derivative $I(\rho) = (1 + \rho^2)/(1 - \rho^2)^2$ at $q = \rho$. Also,

$$\rho_{h,j} = \rho_j - \kappa_j s_\nu(h)R_j\mathcal{M}_\nu'(\alpha R_j) + o\{s_\nu(h)\}.$$

A second-order expansion of (18), followed by completing the weighted square, gives (22) and (23). The positive Hessian $\sum_j \lambda_j$ and target uniqueness at $h = 0$ justify the local minimizer expansion.

At the remaining integer transition,

$$\mathcal{M}_2(x) = 1 - \frac{x^2}{4} + x^4\left\{\frac{3}{64} - \frac{\log(x/2) + \gamma_{\mathrm{E}}}{16}\right\} + O\{x^6(1 + |\log x|)\}.$$

For fixed $\nu > 2$, four-times differentiability yields $\mathcal{M}_\nu(x) = 1 - x^2/\{4(\nu - 1)\} + O(x^4)$. These formulas, and the fractional term for $1 < \nu < 2$, give the three stated smooth-regime remainders after division and inverse expansion.

4 FINITE-GRID SUPPORT-AWARE LIKELIHOOD

The theorem covers translation-invariant interior averaging. For a finite simulation, let X_{in} denote raw sites and let $S_h \in \mathbb{R}^{m\times n}$ be a rectangular row-stochastic observation operator. With noise present before smoothing,

$$\mathbf{y} = S_h\{Y(X_{\mathrm{in}}) + \boldsymbol{\varepsilon}\}, \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}_n(0, \tau^2 I_n), \tag{31}$$

the support-aware covariance is

$$\Sigma_{\mathrm{corr}}(\alpha) = S_h\{K_\alpha(X_{\mathrm{in}}, X_{\mathrm{in}}) + \tau^2 I_n\}S_h^\top. \tag{32}$$

The naive declared-center point-support covariance is instead

$$\Sigma_{\mathrm{naive}}(\alpha) = K_\alpha(X_{\mathrm{out}}, X_{\mathrm{out}}) + \tau^2 I_m. \tag{33}$$

Besides ignoring support, (33) treats pre-smoothing noise as independent post-smoothing noise.

With variance, smoothness, and nugget fixed, the finite-design KL target minimizes

$$Q_n(\alpha) = \frac{1}{2}\left[\log\det\Sigma(\alpha) + \mathrm{tr}\{\Sigma(\alpha)^{-1}\Sigma_0\}\right]. \tag{34}$$

The multi-lag result has an exact full-Gaussian analogue on any fixed interior site design. Let $x_1, \ldots, x_p$ be distinct, let $\theta = (\log\sigma^2, \log\alpha)^\top$, and write $\Sigma(\theta)$ for the point-support Matérn covariance at those sites with fixed ν. Entrywise application of the zero- and nonzero-lag expansions gives

$$\Sigma_h = \Sigma_0 + s_\nu(h)\Gamma_\nu + o\{s_\nu(h)\}. \tag{35}$$

Put $P = \Sigma_0^{-1}$, define the Fisher inner product

$$\langle A, B\rangle_P := \tfrac{1}{2}\,\mathrm{tr}(PAPB), \tag{36}$$

and let $\dot\Sigma_j = \partial_j\Sigma(\theta_0)$. Define

$$\mathcal{J}_{jk} = \langle\dot\Sigma_j, \dot\Sigma_k\rangle_P, \qquad g_{\nu,j} = \langle\dot\Sigma_j, \Gamma_\nu\rangle_P. \tag{37}$$

Proposition 4 (Finite-design full-likelihood projection). *Suppose θ_0 is interior to a compact parameter set, $\mathcal{J}$ is nonsingular, and the KL target $\theta_h^\dagger$ is the local minimizer converging to θ_0. Let $K_h^\star$ denote the minimum full Gaussian KL criterion. With $\beta_\nu = \mathcal{J}^{-1}g_\nu$,*

$$\theta_h^\dagger - \theta_0 = \beta_\nu s_\nu(h) + o\{s_\nu(h)\}, \tag{38}$$

$$K_h^\star = \tfrac{1}{4}\,\mathrm{tr}(PE_\nu PE_\nu)s_\nu(h)^2 + o\{s_\nu(h)^2\}, \tag{39}$$

where $E_\nu = \Gamma_\nu - \sum_{j=1}^2\beta_{\nu,j}\dot\Sigma_j$ satisfies $\langle\dot\Sigma_j, E_\nu\rangle_P = 0$ for $j = 1, 2$. Hence the full likelihood is genuinely misspecified to leading order exactly when the support perturbation has a nonzero component outside the variance–decay covariance tangent space.

To prove Proposition 4, differentiate the Gaussian KL criterion at θ_0. Equations (35) and the matrix inverse derivative give score $-g_\nu s_\nu(h) + o\{s_\nu(h)\}$ and Hessian $\mathcal{J} + o(1)$. The implicit function theorem yields (38). The local Gaussian KL expansion is $\frac{1}{4}\operatorname{tr}\{P(\Gamma_\nu - \sum_j \beta_{\nu,j}\dot{\Sigma}_j)P(\Gamma_\nu - \sum_j \beta_{\nu,j}\dot{\Sigma}_j)\}s_\nu(h)^2$, which proves (39). Nonsingularity of $\mathcal{J}$, rather than increasing- or fixed-domain asymptotics, is the required finite-design identifiability condition.

Unlike Theorem 2, this general projection does not imply a universal sign for the decay coordinate: $\mathcal{J}^{-1}g_\nu$ depends on the complete site design. The selected 3×3 regular-grid audit below has range inflation for all four tested smoothnesses, but that fact is numerical, not an extra theorem.

The corrected family contains Σ_0 when the operator and fixed nuisance parameters are correctly specified and decay is identifiable; the naive family generally does not. Our primary experiment sets $\tau = 0$, isolating field support from noise-timing misspecification. A separate $\tau^2 = 0.05$ run is retained only as a combined-misspecification artifact. Rectangular S_h is intentional: an invertible square, parameter-independent transform may merely change coordinates, whereas aggregation reduces raw sites to fewer supported outputs.

Boundary row normalization is outside the interior theorem. If two location-dependent scaled kernels have means μ_s and μ_t, then

$$C_{h,s,t}(r) = C(r) + h\nabla C(r)^\top(\mu_s - \mu_t) + O(h^2).$$

A truncated kernel can therefore create a first-order term and destroy stationarity; the interior theorem gives no boundary sign guarantee. We include one boundary and one fixed irregular design as illustrations, not as a boundary scaling study.

4.1 Finite-library high-dimensional certificate

The growing-p benchmark jointly varies variance and decay on a deterministic finite grid. The following standard quadratic-form consequence explains which part of its error can shrink with information; it is supporting machinery, not a new concentration inequality.

Proposition 5 (Gaussian finite-library certificate). *Let $0 < \delta < 1$, $\Sigma_0 \in \mathbb{S}^p_{++}$, and $X_1, \ldots, X_N \overset{\text{iid}}{\sim} \mathcal{N}_p(0, \Sigma_0)$. Let $\{\Sigma_\theta : \theta \in \mathcal{G}\}$ be a deterministic library of M covariances in $\mathbb{S}^p_{++}$. Define*

$$\widehat{\Sigma}_N = N^{-1}\sum_{i=1}^{N} X_i X_i^\top, \tag{40}$$

$$\widehat{L}_N(\theta) = \frac{1}{2p}\left\{\log\det\Sigma_\theta + \operatorname{tr}(\Sigma_\theta^{-1}\widehat{\Sigma}_N)\right\}, \tag{41}$$

$$L(\theta) = \mathbb{E}\widehat{L}_N(\theta), \qquad A_\theta = \Sigma_0^{1/2}\Sigma_\theta^{-1}\Sigma_0^{1/2}. \tag{42}$$

For $t = \log(2M/\delta)$, with probability at least $1 - \delta$, simultaneously for every $\theta \in \mathcal{G}$,

$$|\widehat{L}_N(\theta) - L(\theta)| \le \frac{\|A_\theta\|_{\mathrm{F}}}{p}\sqrt{\frac{t}{N}} + \frac{\|A_\theta\|_{\mathrm{op}}}{p}\frac{t}{N} =: u_\theta. \tag{43}$$

If $\widehat{\theta}$ and $\theta^\star$ minimize the empirical and population criteria, respectively, then

$$L(\widehat{\theta}) - L(\theta^\star) \le 2\max_{\theta\in\mathcal{G}} u_\theta. \tag{44}$$

Moreover, $L(\theta)$ differs by a candidate-independent constant from

$$p^{-1}D_{\mathrm{KL}}\{\mathcal{N}_p(0,\Sigma_0)\,\|\,\mathcal{N}_p(0,\Sigma_\theta)\}. \tag{45}$$

To prove (43), write $X_i = \Sigma_0^{1/2}Z_i$, stack the standard Gaussian vectors, and apply the Gaussian quadratic-form inequality to $I_N \otimes A_\theta$. Its Frobenius and operator norms are $\sqrt{N}\|A_\theta\|_{\mathrm{F}}$ and $\|A_\theta\|_{\mathrm{op}}$; division by $2Np$ and a union bound give the displayed constants [13, 15]. The ERM result is the usual three-term comparison through $\widehat{L}_N$.

No coordinate independence is used: all within-field spatial dependence is in Σ_0. Writing

$$r_2(A_\theta) = \frac{\|A_\theta\|_{\mathrm{F}}^2}{\|A_\theta\|_{\mathrm{op}}^2},$$

the radius in (43) is

$$\frac{\|A_\theta\|_{\mathrm{op}}}{p}\left\{\sqrt{\frac{r_2(A_\theta)t}{N}} + \frac{t}{N}\right\}.$$

Thus the relevant dimension is spectral: increasing p helps only when the operator norm and effective-rank geometry remain controlled. If $c\Sigma_0 \preceq \Sigma_\theta \preceq C\Sigma_0$ uniformly, then

$$\max_\theta u_\theta \le c^{-1}\left\{\sqrt{\frac{\log(2M/\delta)}{Np}} + \frac{\log(2M/\delta)}{Np}\right\}. \tag{46}$$

This conditional rate does not imply separate fixed-domain range and variance consistency: relative spectra or parameter curvature may deteriorate with p. The benchmark therefore reports the actual matrix norms and the exact candidate-specific radius.

5 SYNTHETIC VALIDATION

Table 2 makes the theorem–experiment contract explicit. Every track declares its controls, a nonrandom oracle or population target, and a falsifiable numerical gate. Methods can therefore be scored against both the physical parameter and their support-specific KL target rather than one ambiguous target. Saved metadata record resolved factors, seeds, environment, and hashes; the verifier rejects incomplete grids, failed gates, or altered paper inputs.

5.1 Continuous phase oracle

We used a two-dimensional product Epanechnikov density, $\alpha = R = 1$, tensor Gauss–Legendre order 96, 18 bandwidths from 0.003 to 0.3, and $\nu \in \{0.25, 0.5, 0.75, 1, 1.5, 2.5\}$. The density of each coordinate of D is available in closed form, so (3) is evaluated by deterministic quadrature. We solved (5) by monotone root finding. For this kernel $L = \sqrt{2}$, so the two largest displayed bandwidths, 0.229 and 0.3, are stress points outside the sufficient Taylor neighborhood $h \le R/(4L)$. All slope and coefficient checks use the six smallest bandwidths, which are inside that neighborhood. Repeating at orders 64 and 128 changed pseudo-decay by at most 4.4×10^{-8} and 5.2×10^{-9}, respectively, relative to order 96. The released robustness audit repeats the coefficient check in dimensions one through three and for a second compact kernel.

All 108 shifts $\alpha - \alpha_h^\dagger$ were positive. For smoothness 0.25, 0.5, and 0.75, log–log slopes over the six smallest bandwidths were 0.515, 0.999, and 1.468, versus theoretical powers 0.5, 1, and 1.5. The raw

Table 2: SupportShift benchmark contract. The release supplies the listed controls, targets, outputs, and machine-readable pass/fail gates.

Track	Nonrandom target	Principal controls	Reported scoring or falsification gate
Continuous phase and transition	Exact pairwise KL pseudo-decay	h, ν, d, kernel family, quadrature order	Sign, local rate and coefficient; kernel–dimension and quadrature stability
Multi-lag and full likelihood	Composite and finite-design Gaussian KL projections	h, ν, αR, lag weights, site design	Shift coefficient and first nonzero minimum-KL term
Directional support	Exact pairwise KL pseudo-decay by lag angle	h, ν, angle, support aspect	h^2 contrast coefficient and quadrature stability
Finite-grid likelihood	Full Gaussian KL minimizer	h, ν, fitted support, boundary, domain size	Model containment, joint nuisance movement, matched-size boundary contrast
Replicated high dimension	Finite-library Gaussian KL minimizer	p, N, ν, fitted support model	Candidatewise concentration, ERM inequality, target and physical RMSE

slope at $\nu = 1$ was 1.817, reflecting the logarithmic modifier. Slopes for $\nu = 1.5$ and 2.5 were 1.994 and 2.000. At the smallest bandwidth, ratios of the exact shift to the full theorem leading term were 1.021, 1.000, 0.958, 1.059, 0.997, and 1.000 in increasing order of ν.

The pointwise qualifier matters at practical bandwidths. We therefore audited the threshold layer on 111 deterministic cells with $\nu \in [0.55, 1.45]$, spacing 0.025, and $h \in \{0.01, 0.02, 0.05\}$. The smallest exact-shift to one-term ratio was 0.154; equivalently, the one-term prediction was up to 6.51 times the exact shift near, but not at, $\nu = 1$. In contrast, the maximum relative error of $\widetilde{\alpha}_h$ was 0.099%, and the maximum relative error in variance loss was 0.084%. Every exact and approximate shift had the theorem-predicted sign. Repeating the exact target at quadrature orders 64 and 128 changed it by at most 4.0×10^{-9} and 3.1×10^{-10}, respectively, relative to the shift computed at order 96. These are finite-grid diagnostics, not a claimed joint uniform asymptotic theorem.

5.2 Genuine likelihood projection audits

We evaluated Theorem 2 at $\alpha R \in \{0.5, 1, 1.5, 2\}$, equal weights, $\nu \in \{0.5, 1, 1.5, 2.5\}$, and $h \in \{0.005, 0.01, 0.02\}$. All 12 composite targets had $\alpha - \alpha_h^{\mathrm{C}} > 0$ and strictly positive minimum KL. At the smallest bandwidth, the maximum relative errors of the leading shift and minimum-KL terms were 1.41% and 9.85%, respectively. The lag coefficients vary substantially—for $\nu = 0.5$, from 1.61 at $\alpha R = 0.5$ to 0.40 at $\alpha R = 2$—so the residual in (23) is not a numerical roundoff effect.

For Proposition 4, we used a 3×3 regular grid with unit spacing and jointly profiled variance and decay. Every one of the 12 exact continuously smoothed targets had positive minimum KL and range inflation. At $h = 0.005$, the maximum relative errors of the predicted decay shift, log-variance shift, and minimum KL were 5.42%, 12.68%, and 1.77%. These signs describe this selected design; only the matrix projection and residual-KL formulas are general.

5.3 Directional support oracle

The directional track uses (28) with $\varrho \in \{1, 4\}$, 19 lag angles from zero to 90 degrees, $\nu \in \{0.5, 1, 1.5, 2.5\}$, and 14 bandwidths from 0.003 to 0.15. All 2,128 inverse-range shifts were positive. At $\varrho = 4$, the major/minor smooth-regime coefficient ratios were 1.833 for $\nu = 1.5$ and 3.250 for $\nu = 2.5$, exactly as predicted by (29). At $h = 0.15$, the inferred range inflation along the major and minor axes was 13.46% and 11.47% for $\nu = 0.5$, and 1.56% and 0.76% for $\nu = 1.5$.

The major-minus-minor shift divided by its $D_\nu h^2$ term differed from one by at most 4.3×10^{-6} at the smallest bandwidth over all four smoothnesses. Relative to order 96, orders 64 and 128 changed any individual shift by at most 1.3×10^{-8} of that shift and changed a directional contrast by at most 1.2×10^{-9} of that contrast.

5.4 Finite-grid full likelihood

The core design uses a 19×19 input lattice with spacing 0.25 and every second site as a candidate output center, trimmed by 0.75 in each coordinate. We set $\sigma^2 = \alpha = 1$, $\tau^2 = 0$, $\nu \in \{0.5, 1, 1.5, 2.5\}$, and $h \in \{0, 0.3, 0.5, 0.7\}$. Radial Epanechnikov rows are normalized exactly. Stress designs retain boundary centers, perturb each coordinate uniformly in $[-0.04, 0.04]$ before clipping, and grow one-dimensional domains from 100 to 400 raw sites. The irregular panel is one fixed realization.

For each configuration, (34) was minimized by bounded scalar optimization over $\log \alpha \in [\log(0.02), \log(5)]$. Two hundred replicate likelihoods were evaluated on a 161-point log-decay profile; interior minima were refined by three-point quadratic interpolation and endpoint minima were retained as boundary fits. Gaussian raw-observation vectors were transformed by S_h. Common random numbers were reused across bandwidths. No adaptive diagonal jitter was added; a Cholesky or population-optimization failure would abort the configuration and make the reducer mark the run incomplete.

The immutable run contained 21 configurations and 8,400 fits. The reducer verified configuration hashes, row counts, and unique model–replicate keys; independent checks covered finiteness, model labels, replicate indices, and metadata. Across the core designs, corrected population targets were within 1.3×10^{-7} of one. At $h = 0.7$, naive targets were 0.152, 0.414, 0.573, and 0.740 as ν increased; Monte Carlo means were 0.151, 0.407, 0.570, and 0.738. Corrected means were 1.090, 0.990, 0.990, and 0.994. The first was within two Monte Carlo standard errors of its target. All six core boundary-hitting fits were retained.

The earlier boundary stress changed output dimension. We therefore added two translated 4×4 blocks on the same 19×19 latent grid: one touches the lower-left boundary and one is interior, while both have $p = 16$ outputs and identical relative geometry. At $h = 0.7$,

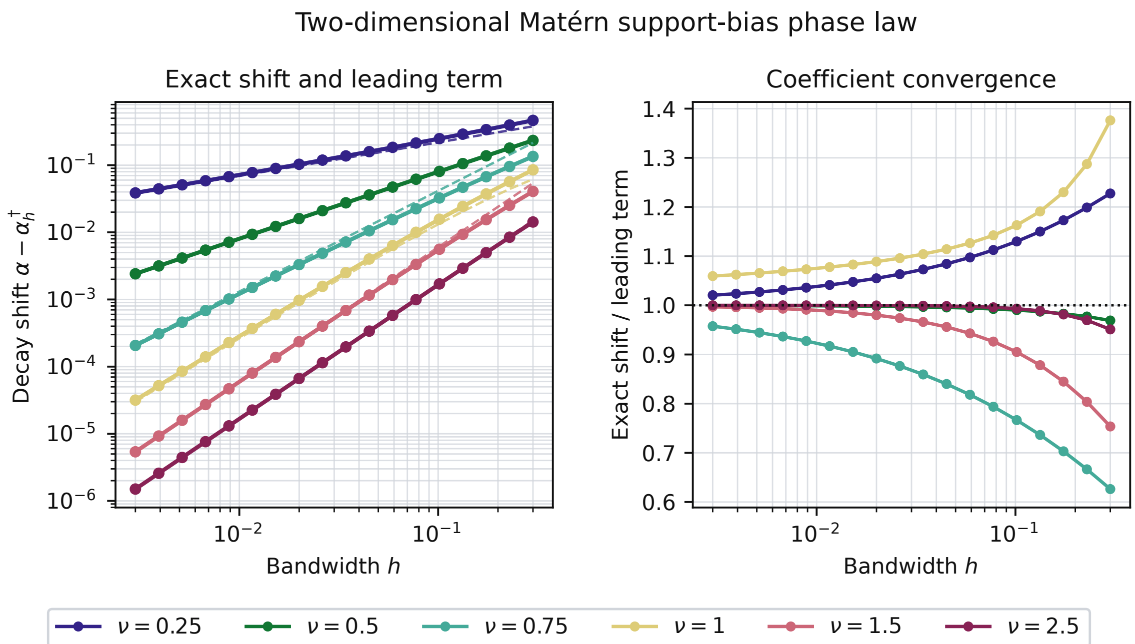

Figure 1: Continuous two-dimensional pseudo-decay shift. Left: exact shifts (solid) and theorem leading terms with explicit coefficients (dashed). Right: their ratio, which approaches one as $h \downarrow 0$.

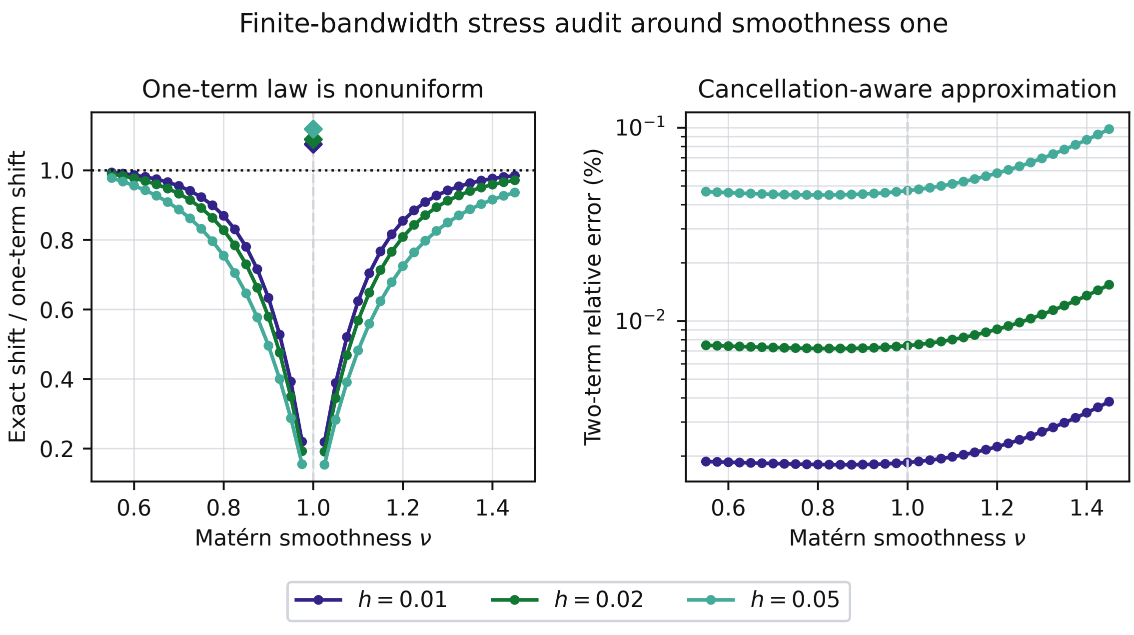

Figure 2: Finite-bandwidth transition stress around $\nu = 1$. Left: the fixed-ν one-term law is accurate away from the crossover but nonuniform near it. Right: retaining the canceling quadratic and fractional terms gives a continuous, sub-0.1% approximation on the audited grid.

their point-support targets differ by 0.030–0.062 across smoothnesses, isolating a boundary effect without a dimension change.

A less-oracular model that uses the correct radial support shape but bandwidth $0.75h$ reduced population KL to 16.7–32.3% of the

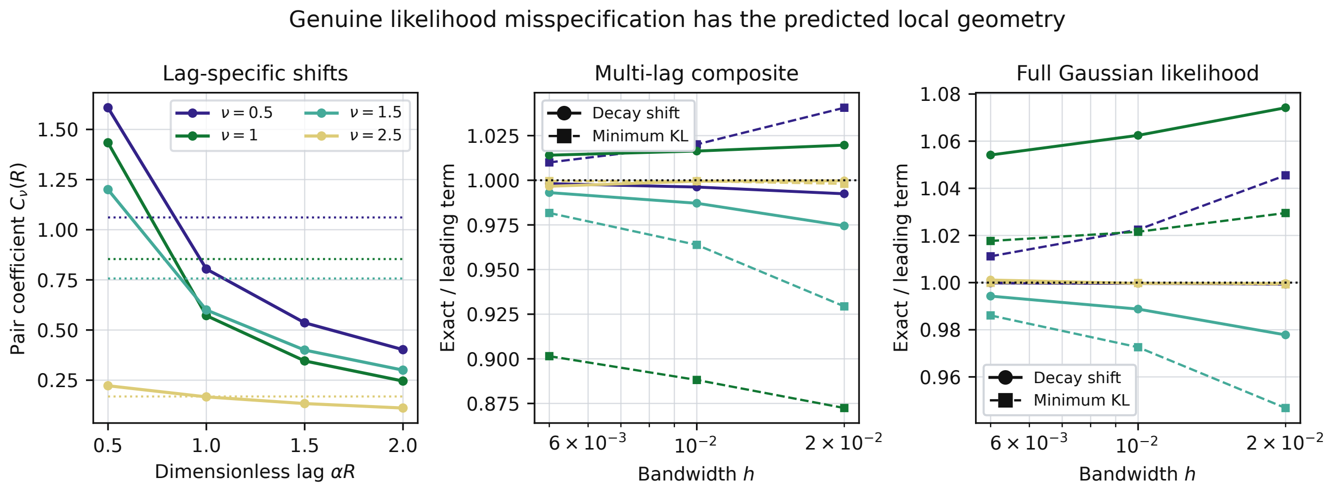

Figure 3: From a saturated pair match to genuine misspecification. Dotted segments in the left panel are the information-weighted composite coefficients. The middle and right panels validate both the pseudo-parameter shift and the first nonzero minimum-KL term.

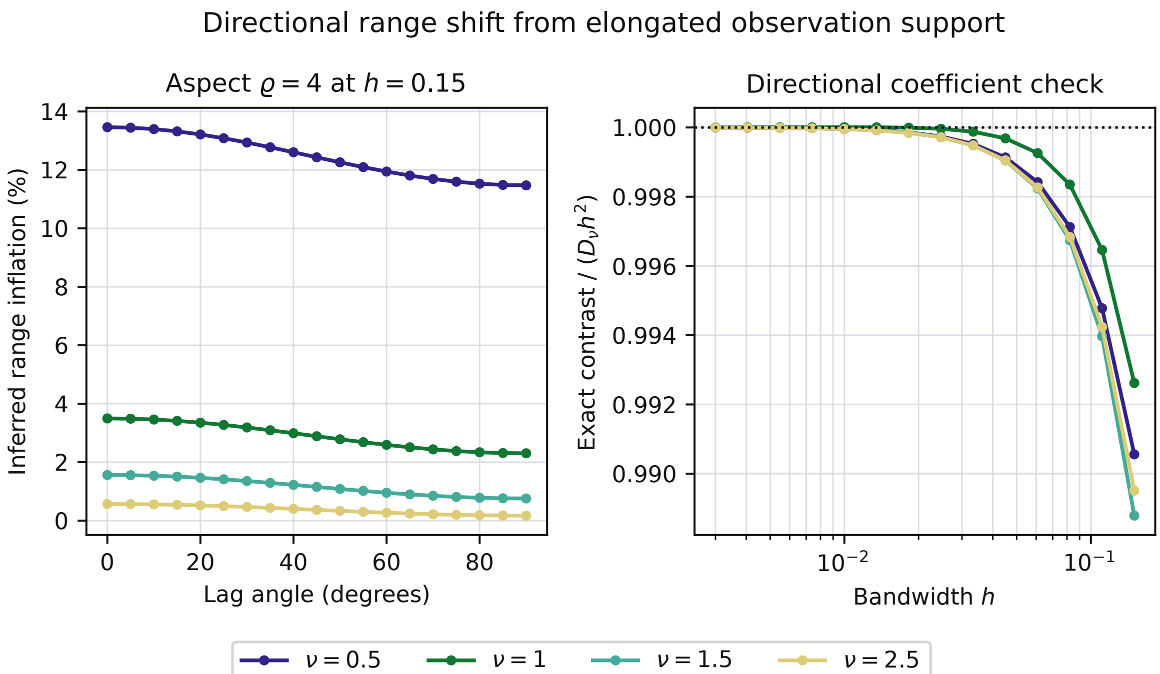

Figure 4: An isotropic latent field observed through elongated support. Left: apparent range inflation versus lag angle at support aspect four and $h = 0.15$. Right: exact major–minor inverse-range contrast divided by $D_\nu h^2$.

point-support KL. The exact-support target recovered the truth to 4.1×10^{-7} in decay. Table 4 reports the matched comparison. For the fixed jittered design the original target moved from 0.289 to 0.284 at $h = 0.5$. In the domain-growth check, increasing raw sites from 100 to 400 reduced corrected-estimator standard deviation from 0.529 to 0.177 and naive-estimator standard deviation from 0.151 to 0.068, while at $n = 400$ their means were within 0.026 and 0.009 of their respective targets.

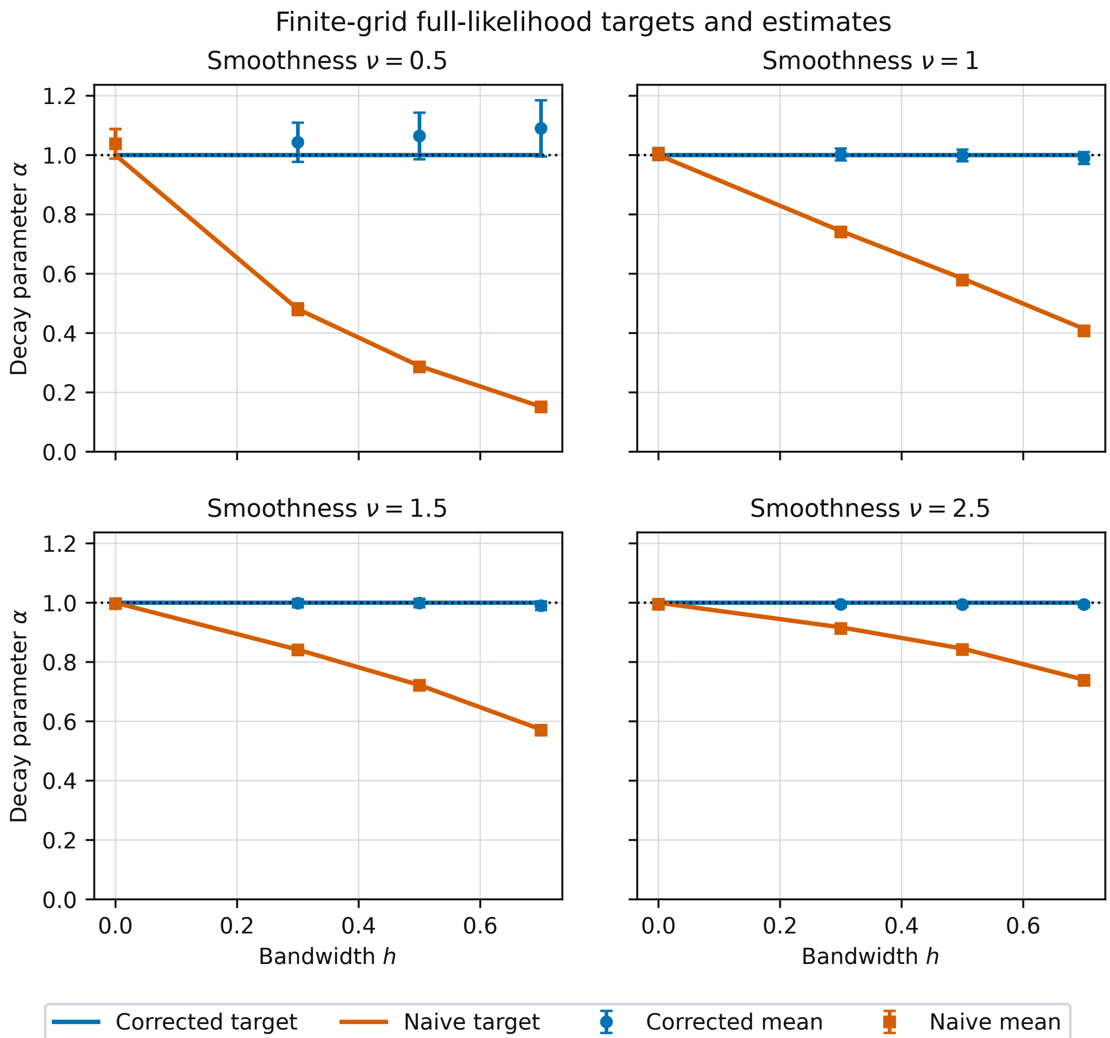

Figure 5: Finite-design full-likelihood targets and Monte Carlo mean estimates. Error bars show two Monte Carlo standard errors of the mean.

Table 3: Finite-design results at $h = 0.7$. MCSE is for the reported mean.

ν	Model	Target	Mean	MCSE	Bound hits
0.5	Corrected	1.000	1.090	0.047	3
0.5	Naive	0.152	0.151	0.004	0
1	Corrected	1.000	0.990	0.010	0
1	Naive	0.414	0.407	0.005	0
1.5	Corrected	1.000	0.990	0.007	0
1.5	Naive	0.573	0.570	0.005	0
2.5	Corrected	1.000	0.994	0.004	0
2.5	Naive	0.740	0.738	0.003	0

5.5 Joint smoothness–decay and partial-support fits

To test the known-smoothness restriction, we jointly selected (ν, α, σ^2) from a fixed library with 18 smoothness values, 52 log-spaced decay values, and analytically profiled variance. The data use an 11×11 latent grid, 16 interior outputs, true $\nu \in \{0.5, 1, 1.5, 2.5\}$, $h \in \{0.35, 0.5\}$, $N = 16$ independent fields, and 100 trials per cell. We compared exact support, $0.75h$ support, and point support, producing 2,400 fits.

The exact-support population target equaled the physical parameter in all eight design cells. Point support changed the population smoothness target in all eight, always upward: for example, at $h = 0.5$ the target moved from $(\nu, \alpha) = (0.5, 1)$ to $(2, 2.176)$, and from $(2.5, 1)$ to $(3.5, 1.338)$. Thus the fixed-ν range-inflation theorem does not describe the joint fit: smoothness absorbs enough

Table 4: Matched-size boundary comparison at $h = 0.7$. I and B denote interior and boundary blocks. The KL columns report the intermediate $0.75h$-support model relative to the point-support model; smaller is better.

ν	$\alpha^*_{\mathrm{point,I}}$	$\alpha^*_{\mathrm{point,B}}$	$\mathrm{KL}_{.75h}/\mathrm{KL}_{\mathrm{point,I}}$	$\mathrm{KL}_{.75h}/\mathrm{KL}_{\mathrm{point,B}}$	p
0.5	0.225	0.183	0.167	0.234	16
1	0.391	0.337	0.187	0.260	16
1.5	0.525	0.495	0.202	0.281	16
2.5	0.709	0.771	0.214	0.323	16

aggregation that the fitted inverse range exceeds one in all eight point-support cells. The partial-support population criterion improved on point support in all eight cells, providing a nontrivial method ordering rather than only exact-versus-naive containment. Monte Carlo medians generally tracked their discrete targets; the released aggregate table contains every population target and median.

5.6 Replicated-field high-dimensional track

For $q \in \{4, 6, 8, 10\}$, a padded latent lattice with spacing 0.25 was mapped by a full-row-rank radial Epanechnikov operator with $h = 0.5$ to a $q \times q$ output lattice, giving $p \in \{16, 36, 64, 100\}$. We used $\nu \in \{0.5, 1.5\}$, $\sigma^2 = \alpha = 1$, and $N \in \{1, 4, 16, 64\}$ independent fields. Corrected candidates were $\sigma^2 S_h R_\alpha S_h^\top$; naive candidates were $\sigma^2 R_\alpha$ at output centers. The deterministic $101 \times 161 = 16{,}261$ variance–decay library spans $[0.35, 2.5] \times [0.15, 1.6]$ and contains $(1, 1)$. This truth anchoring is an oracle benchmark control, not a deployable grid-selection rule; the complete library was fixed before the final Monte Carlo run.

Each (p, ν) cell used 200 nested replicate-field sequences, giving 64 model–design–sample-size cells and 12,800 exact finite-library fits. The discrete population objective was within 2.63×10^{-5} per coordinate of the continuously profiled objective in all 16 design–model cases, below the 5×10^{-5} tolerance fixed before simulation. The corresponding maximum continuous-to-library parameter distances were 0.00499 in decay and 0.0297 in variance for point support, and below 4.1×10^{-7} for both coordinates under exact support. Thus the reported parameter floor is explicit, not hidden behind an objective-only audit. No adaptive jitter or coordinate-independence approximation was used; the declared support operator is applied directly to every latent field.

With $\delta = 0.05$, every trial in every cell satisfied the actual candidatewise simultaneous event (43). Across all fits,

$$\max_\theta \frac{|\widehat{L}_N(\theta) - L(\theta)|}{u_\theta} \le 0.789.$$

Each certificate applies separately to its fixed cell, not jointly across all 64. The deterministic ERM inequality (44) also held for every fit. Across cells, the 95th percentile of the worst criterion deviation was at most 0.402 times the worst-candidate radius, with median ratio 0.298. For each fixed model, p, and ν, its log–log slope against N ranged from -0.528 to -0.454, with median -0.505, matching the leading $N^{-1/2}$ dependence. At fixed N, regressing the same 95th-percentile deviation on p gave slopes from -0.509 to -0.297, with median -0.413. The departure from a universal $-1/2$ is expected: Proposition 5 depends on relative operator norms and effective rank, not coordinate count alone.

At $p = 100$, $\nu = 1.5$, the naive finite-grid decay target was 0.594 (continuous target 0.589), versus the physical value one. From $N = 1$ to 64, naive RMSE to its own target fell from 0.124 to 0.015, while its RMSE to the physical value was 0.409 at $N = 64$; the support-aware RMSE fell from 0.206 to 0.024. At $\nu = 0.5, N = 64$, naive RMSE was 0.015 to its target but 0.681 to the physical value. Thus increasing information reduced sampling error without repairing support misspecification. These parameter summaries are empirical; the proposition controls criterion error and excess risk unless a curvature condition is added.

6 DISCUSSION

Propagating covariance through known support is classical; our object is the parameter selected when that propagation is omitted. Unlike finite range-of-influence rules [6], the Matérn KL target moves at a smoothness-dependent power: an origin cusp drives the rough-field fractional rate, while a Bessel identity fixes the smooth-field sign.

The simulations compute finite-design targets and separate misspecification from sampling error; they do not replace the proofs. The scope is deliberately narrow: the universal sign fixes smoothness, the multi-lag theorem concerns a declared pair composite, and the full-likelihood coefficient is design specific. Boundary normalization may create first-order terms, joint nuisance parameters may be weakly identified, and the probability radius is an oracle benchmark certificate. Fixed-resolution smoothing also does not converge to continuous convolution merely through domain growth.

Fuentes's block-average spectral likelihood and small-pixel approximation is the closest analytic setup we located [9]; recent finite-grid Matérn work instead studies discretization and edge effects [17]. Neither derives this omitted-support KL phase law. This is a scoped comparison, not a claim of exhaustive priority.

7 CONCLUSION

Ignored interior support shifts the fixed-smoothness inverse range by $h^{2\nu}$, $h^2 \log(1/h)$, or h^2. Multi-lag and finite-design projections expose genuine residual KL, while the replicated-field certificate separates likelihood noise from population displacement. Joint smoothness can reverse the fixed-smoothness decay direction; no universal sign is asserted for boundary support or arbitrary full-likelihood fits.

Code, manifests, aggregate data, and a one-command numerical-claim verifier are available in release v1.3.2.

REFERENCES

[1] François Bachoc. 2018. Asymptotic Analysis of Covariance Parameter Estimation for Gaussian Processes in the Misspecified Case. *Bernoulli* 24, 2 (2018), 1531–1575. https://doi.org/10.3150/16-BEJ906

[2] Claude Bellehumeur and Pierre Legendre. 1997. Aggregation of Sampling Units: An Analytical Solution to Predict Variance. *Geographical Analysis* 29, 3 (1997), 258–266. https://doi.org/10.1111/j.1538-4632.1997.tb00961.x

[3] Moreno Bevilacqua, Carlo Gaetan, Jorge Mateu, and Emilio Porcu. 2012. Estimating Space and Space-Time Covariance Functions for Large Data Sets: A Weighted Composite Likelihood Approach. *J. Amer. Statist. Assoc.* 107, 497 (2012), 268–280. https://doi.org/10.1080/01621459.2011.646928

[4] Erick A. Chacón-Montalván, Peter M. Atkinson, Christopher Nemeth, Benjamin M. Taylor, and Paula Moraga. 2024. Spatial Latent Gaussian Modelling with Change of Support. https://doi.org/10.48550/arXiv.2403.08514 arXiv:2403.08514 [stat.ME]

[5] Jean-Paul Chilès and Pierre Delfiner. 2012. *Geostatistics: Modeling Spatial Uncertainty* (2 ed.). Wiley, Hoboken, NJ. https://doi.org/10.1002/9781118136188

[6] Isobel Clark. 1977. Regularization of a Semivariogram. *Computers & Geosciences* 3, 2 (1977), 341–346. https://doi.org/10.1016/0098-3004(77)90010-3

[7] David Clifford, Alex B. McBratney, James Taylor, and Brett M. Whelan. 2006. Generalized Analysis of Spatial Variation in Yield Monitor Data. *The Journal of Agricultural Science* 144, 1 (2006), 45–51. https://doi.org/10.1017/S0021859606005892

[8] Victor De Oliveira and Zifei Han. 2022. On Information About Covariance Parameters in Gaussian Matérn Random Fields. *Journal of Agricultural, Biological and Environmental Statistics* 27, 4 (2022), 690–712. https://doi.org/10.1007/s13253-022-00510-5

[9] Montserrat Fuentes. 2007. Approximate Likelihood for Large Irregularly Spaced Spatial Data. *J. Amer. Statist. Assoc.* 102, 477 (2007), 321–331. https://doi.org/10.1198/016214506000000852

[10] Alan E. Gelfand, Li Zhu, and Bradley P. Carlin. 2001. On the Change of Support Problem for Spatio-Temporal Data. *Biostatistics* 2, 1 (2001), 31–45. https://doi.org/10.1093/biostatistics/2.1.31

[11] Carol A. Gotway and Linda J. Young. 2002. Combining Incompatible Spatial Data. *J. Amer. Statist. Assoc.* 97, 458 (2002), 632–648. https://doi.org/10.1198/016214502760047140

[12] Matthew J. Heaton, Abhirup Datta, Andrew O. Finley, Reinhard Furrer, Joseph Guinness, Rajarshi Guhaniyogi, Florian Gerber, Robert B. Gramacy, Dorit Hammerling, Matthias Katzfuss, Finn Lindgren, Douglas W. Nychka, Furong Sun, and Andrew Zammit-Mangion. 2019. A Case Study Competition Among Methods for Analyzing Large Spatial Data. *Journal of Agricultural, Biological and Environmental Statistics* 24, 3 (2019), 398–425. https://doi.org/10.1007/s13253-018-00348-w

[13] Daniel Hsu, Sham M. Kakade, and Tong Zhang. 2012. A Tail Inequality for Quadratic Forms of Subgaussian Random Vectors. *Electronic Communications in Probability* 17 (2012), 1–6. https://doi.org/10.1214/ECP.v17-2079

[14] Cari G. Kaufman and Benjamin A. Shaby. 2013. The Role of the Range Parameter for Estimation and Prediction in Geostatistics. *Biometrika* 100, 2 (2013), 473–484. https://doi.org/10.1093/biomet/ass079

[15] Béatrice Laurent and Pascal Massart. 2000. Adaptive Estimation of a Quadratic Functional by Model Selection. *The Annals of Statistics* 28, 5 (2000), 1302–1338. https://doi.org/10.1214/aos/1015957395

[16] Christopher J. Paciorek and Mark J. Schervish. 2006. Spatial Modelling Using a New Class of Nonstationary Covariance Functions. *Environmetrics* 17, 5 (2006), 483–506. https://doi.org/10.1002/env.785

[17] Frederik J. Simons, Olivia L. Walbert, Arthur P. Guillaumin, Gabriel L. Eggers, Kevin W. Lewis, and Sofia C. Olhede. 2026. Maximum-Likelihood Estimation of the Matérn Covariance Structure of Isotropic Spatial Random Fields on Finite, Sampled Grids. *Geophysical Journal International* 245, 2 (2026), ggag044. https://doi.org/10.1093/gji/ggag044

[18] Yusuke Tanaka, Toshiyuki Tanaka, Tomoharu Iwata, Takeshi Kurashima, Maya Okawa, Yasunori Akagi, and Hiroyuki Toda. 2019. Spatially Aggregated Gaussian Processes with Multivariate Areal Outputs. In *Advances in Neural Information Processing Systems*, Vol. 32. Curran Associates, Inc., Red Hook, NY, 12 pages. https://proceedings.neurips.cc/paper/2019/hash/a941493eeea57ede8214fd77d41806bc-Abstract.html

[19] Cristiano Varin, Nancy Reid, and David Firth. 2011. An Overview of Composite Likelihood Methods. *Statistica Sinica* 21, 1 (2011), 5–42.

[20] Hao Zhang. 2004. Inconsistent Estimation and Asymptotically Equal Interpolations in Model-Based Geostatistics. *J. Amer. Statist. Assoc.* 99, 465 (2004), 250–261. https://doi.org/10.1198/016214504000000241

[21] Weiyue Zheng, Andrew Elliott, Claire Miller, and Marian Scott. 2026. A Bayesian INLA–SPDE Approach to Spatio-Temporal Point-Grid Fusion with Change-of-Support and Misaligned Covariates. *Spatial Statistics* 74 (2026), 100998. https://doi.org/10.1016/j.spasta.2026.100998

A PROOF DETAILS FOR THE ANALYTIC CLAIMS

Proof of Theorem 1. Set $f(x) = C(x)/\sigma^2$. Its first four derivatives are uniformly bounded on the declared annulus. Since $D = U - V$ is centrally symmetric and $\mathbb{E}(DD^\top) = 2\Sigma_k$, Taylor's theorem gives

$$\mathbb{E}f(r + hD) = f(r) + \frac{h^2}{2}\operatorname{tr}\{\nabla^2 f(r)\mathbb{E}(DD^\top)\} + O(h^4). \tag{47}$$

For $r = Re$, the radial Hessian is

$$\nabla^2 f(r) = \alpha^2 \mathcal{M}_\nu''(z)ee^\top + \alpha^2\frac{\mathcal{M}_\nu'(z)}{z}(I - ee^\top),$$

which proves Proposition 1.

For noninteger ν, the convergent origin expansion is

$$\mathcal{M}_\nu(x) = \sum_{j\geq 0}\frac{(x^2/4)^j}{j!(1-\nu)_j} + b_\nu x^{2\nu}\sum_{j\geq 0}\frac{(x^2/4)^j}{j!(1+\nu)_j}, \quad b_\nu = \frac{\Gamma(-\nu)}{2^{2\nu}\Gamma(\nu)}. \tag{48}$$

Because D is bounded, termwise expectation at $x = \alpha h\|D\|$ is dominated uniformly for α in a compact subset of $(0,\infty)$. For $0 < \nu < 1$, the fractional term is leading and the next terms are $O(h^2)$. For $1 < \nu < 2$, the quadratic term is leading and the fractional remainder is $O(h^{2\nu})$. For fixed $\nu > 2$, four-times differentiability at zero gives an $O(h^4)$ remainder. At the two integer transitions,

$$\mathcal{M}_1(x) = 1 + \frac{x^2}{2}\{\log(x/2) + \gamma_{\mathrm{E}} - 1/2\} + O\{x^4(1 + |\log x|)\}, \tag{49}$$

$$\mathcal{M}_2(x) = 1 - \frac{x^2}{4} - \frac{x^4}{16}\{\log(x/2) + \gamma_{\mathrm{E}} - 3/4\} + O\{x^6(1 + |\log x|)\}. \tag{50}$$

These formulas prove (9)–(11), including the stated remainders.

Write $d_h = C_h(0)/\sigma^2$. Dividing (47) by d_h, then applying the local inverse of $\alpha \mapsto \mathcal{M}_\nu(\alpha R)$, is valid because

$$\mathcal{M}_\nu'(z) = -\frac{2^{1-\nu}}{\Gamma(\nu)}z^\nu K_{\nu-1}(z) < 0.$$

This gives (12)–(14). In the smooth regime, the order-h^2 normalized-correlation coefficient is

$$B_{\nu,k}(r) + \mathcal{M}_\nu(z)\frac{\alpha^2 m_2}{4(\nu-1)}.$$

Using $m_2 = 2T_k$, the Matérn differential equation, and $K_\nu(z) = K_{\nu-2}(z) + 2(\nu-1)K_{\nu-1}(z)/z$ reduces it to the displayed identity $A_{\nu,k}(r) > 0$. Division by the negative $R\mathcal{M}_\nu'(z)$ proves that $\alpha_h^\dagger - \alpha < 0$. □

Proof of Theorem 2. For $|\rho|, |q| < 1$, put $P_\rho = \mathcal{N}(0, \mathrm{R}(\rho))$. Direct evaluation gives

$$d(\rho, q) := D_{\mathrm{KL}}(P_\rho \,\|\, P_q) = \frac{1}{2}\left[\log\frac{1-q^2}{1-\rho^2} - 2 + \frac{2(1-q\rho)}{1-q^2}\right]. \tag{51}$$

At $q = \rho$, $d = \partial_q d = 0$ and $\partial_q^2 d = I(\rho) = (1+\rho^2)/(1-\rho^2)^2$. Theorem 1 and differentiability of the Matérn correlation at positive lags give, uniformly over the finite lag set,

$$\begin{aligned}\rho_{h,j} &= \mathcal{M}_\nu\{(\alpha - \kappa_j s_\nu(h) + o\{s_\nu(h)\})R_j\} \\ &= \rho_j - \kappa_j\dot\rho_j s_\nu(h) + o\{s_\nu(h)\}, \qquad \dot\rho_j = R_j\mathcal{M}_\nu'(\alpha R_j).\end{aligned}$$

For $\eta = \alpha - \gamma s_\nu(h)$, the candidate correlation is $\rho_j - \gamma\dot\rho_j s_\nu(h) + o\{s_\nu(h)\}$. Taylor expansion of (51) therefore yields

$$Q_h^{\mathrm{C}}(\eta) = \frac{s_\nu(h)^2}{2}\sum_j \lambda_j(\kappa_j - \gamma)^2 + o\{s_\nu(h)^2\}.$$

The criterion at $h = 0$ is uniquely minimized by $\eta = \alpha$, and its second derivative there is $\sum_j \lambda_j > 0$. Compactness and uniform convergence place every global minimizer near α; the implicit function theorem then gives the displayed local expansion. Minimizing the quadratic in γ gives $\gamma = \bar\kappa$ and proves both claims. □

Proof of Proposition 4. Let

$$K_h(\theta) = \frac{1}{2}\left[\operatorname{tr}\{\Sigma(\theta)^{-1}\Sigma_h\} - p + \log\det\Sigma(\theta) - \log\det\Sigma_h\right].$$

Using $\partial\Sigma^{-1} = -\Sigma^{-1}(\partial\Sigma)\Sigma^{-1}$, the score at θ_0 is

$$\partial_j K_h(\theta_0) = -s_\nu(h)\langle\dot\Sigma_j, \Gamma_\nu\rangle_P + o\{s_\nu(h)\} = -g_{\nu,j}s_\nu(h) + o\{s_\nu(h)\}.$$

At $h = 0$, the Hessian is the Fisher information $\mathcal{J}$. Nonsingularity and the implicit function theorem give $\theta_h^\dagger - \theta_0 = \mathcal{J}^{-1}g_\nu s_\nu(h) + o\{s_\nu(h)\}$. Finally, for a covariance perturbation H, Gaussian KL has quadratic term $\frac{1}{4}\operatorname{tr}(PHPH)$. Substitution of $H = s_\nu(h)(\Gamma_\nu - \sum_j \beta_{\nu,j}\dot\Sigma_j) + o\{s_\nu(h)\}$ proves (39). The coefficient vanishes exactly when $E_\nu = 0$, because $\operatorname{tr}(PE_\nu PE_\nu) = \|P^{1/2}E_\nu P^{1/2}\|_{\mathrm{F}}^2$. □

Proof of Propositions 2–3. Retaining both displayed terms of (48) gives

$$d_h = 1 + W_{\nu,k}(h) + \begin{cases} O(h^{2\nu+2}), & 0 < \nu < 1, \\ O(h^4), & 1 < \nu < 2. \end{cases}$$

At $\nu = 1$, the corresponding error is $O\{h^4\log(1/h)\}$. Combining these denominators with (47) and using the bounded derivative of the inverse Matérn map proves (26). For $\nu = 1 + \varepsilon$,

$$b_{1+\varepsilon} = \frac{1}{4\varepsilon} + \frac{2\gamma_{\mathrm{E}} - 1 - 2\log 2}{4} + O(\varepsilon), \tag{52}$$

$$m_{2+2\varepsilon}(\alpha h)^{2+2\varepsilon} = (\alpha h)^2\Big[m_2 + 2\varepsilon\{\ell_{2,k} + m_2\log(\alpha h)\} + O(\varepsilon^2)\Big]. \tag{53}$$

The pole cancels the analytic quadratic term, leaving (24) at $\nu = 1$.

For directions e_1, e_2, the common d_h cancels and

$$\rho_h(Re_1) - \rho_h(Re_2) = h^2\{B_{\nu,k}(Re_1) - B_{\nu,k}(Re_2)\} + o(h^2).$$

The inverse-function theorem and

$$\mathcal{M}_\nu''(z) - \mathcal{M}_\nu'(z)/z = 2^{1-\nu}z^\nu K_{\nu-2}(z)/\Gamma(\nu)$$

then give (27) for every fixed $\nu > 0$. □

B PROOF OF THE FINITE-LIBRARY CERTIFICATE

Proof of Proposition 5. Write $X_i = \Sigma_0^{1/2}Z_i$, stack the independent $Z_i \sim \mathcal{N}_p(0, I_p)$, and put $A_\theta = \Sigma_0^{1/2}\Sigma_\theta^{-1}\Sigma_0^{1/2}$. The random likelihood term is the quadratic form associated with $I_N \otimes A_\theta$. The two-sided Gaussian quadratic-form inequality gives, for $t > 0$,

$$\begin{aligned}&\Pr\left(\left|\sum_{i=1}^N X_i^\top\Sigma_\theta^{-1}X_i - N\operatorname{tr}(A_\theta)\right| > 2\sqrt{N}\|A_\theta\|_{\mathrm{F}}\sqrt{t} + 2\|A_\theta\|_{\mathrm{op}}t\right) \\ &\quad\leq 2e^{-t}.\end{aligned} \tag{54}$$

Indeed, the Frobenius and operator norms of the Kronecker matrix are $\sqrt{N}\|A_\theta\|_{\mathrm{F}}$ and $\|A_\theta\|_{\mathrm{op}}$. The log determinant is deterministic. Divide by $2Np$, take $t = \log(2M/\delta)$, and union bound over the deterministic library to obtain (43). On that simultaneous event, the empirical-minimizer comparison through $\widehat{L}_N$ gives

$$L(\widehat{\theta}) - L(\theta^\star) \le [L - \widehat{L}_N](\widehat{\theta}) + [\widehat{L}_N - L](\theta^\star) \le 2\max_\theta u_\theta.$$

Finally, the Gaussian KL formula differs from $pL(\theta)$ only by terms independent of θ. If $c\Sigma_0 \preceq \Sigma_\theta$, congruence after inversion gives $A_\theta \preceq c^{-1}I_p$, hence $\|A_\theta\|_{\mathrm{op}} \le c^{-1}$ and $\|A_\theta\|_{\mathrm{F}} \le \sqrt{p}/c$, which proves (46). □

C DIMENSION–KERNEL ROBUSTNESS AUDIT

As a predeclared implementation audit, we repeated the coefficient check for dimensions $d \in \{1, 2, 3\}$, product Epanechnikov and uniform kernels, four smoothness values, and $h \in \{0.002, 0.004, 0.008\}$. Tensor quadrature used order 48 and was repeated at orders 32 and 64. Table 5 reports the smallest-bandwidth ratio of the exact shift to its leading term.

Table 5: Coefficient-ratio ranges in the 72-cell robustness audit.

Kernel	d	Minimum ratio	Maximum ratio
Epanechnikov	1	0.999	1.169
Epanechnikov	2	0.998	1.056
Epanechnikov	3	0.997	1.005
Uniform	1	0.998	1.130
Uniform	2	0.997	1.016
Uniform	3	0.966	1.000

All 72 shifts were positive. The largest coefficient-ratio error was 0.169, below its 0.20 gate, and the largest quadrature-refinement change was 3.92×10^{-7}, below its 2×10^{-4} gate. This finite audit is not a uniform guarantee over kernels or dimensions.

D ARTIFACT-LEVEL REPRODUCTION

The isolated artifact entry point is `supportshift/README.md`. From a clean checkout, its release-verification command checks the environment and provenance records, hashes every promoted input and paper artifact, validates all declared factor grids and schemas, reconstructs the finite-grid summary from fit-level records, and evaluates the machine-readable claim ledger. A successful full audit reports 12,800 replicated-field fits, 8,400 finite-grid fits, 64 candidatewise coverage cells, 23 hashed source inputs, 33 hashed paper artifacts, and 168 passed numerical claims.

The same entry point gives a small custom-generation command for `run_supportshift_highdim.py`. Users can vary lattice side, independent field count, Matérn smoothness, support bandwidth, and the variance–decay candidate library. Each generated row distinguishes the physical parameter, the model-specific finite-grid KL target, and the random estimate, and the sibling JSON records the complete arguments, derived seeds, dependency versions, Git state, output hash, and validation gates. The deterministic phase, multi-lag, full-likelihood, and directional drivers are listed there as theorem oracles rather than Monte Carlo substitutes.